User Evaluation

Cohort 3 Team 1: Pixels of Promise

Movitz Aaron, Jake Adams, Oliver Belam, Anna Burt, Emily Foran, Sky Haines-Bass, Job Young

University of York

ENG1: Software & Systems Engineering

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1 Method for User Evaluations

1. Recruit some participants

Our approach to recruiting users aimed to ensure a diverse and representative pool of participants, 4 of which being female and 3 being male. The only constraint was that participants had to be a student attending the University of York. To recruit participants, each team member was tasked in finding one to two potential participants who were available during the timeframe we had allocated for the user evaluations. Though we only required seven users, we wanted to account for last-minute cancellations and unavailability hence the large pool of participants. This approach was intended to reduce the risk of delays, increasing the bus factor. All participants filled in a consent form as per our ethics procedures prior to their user evaluation.

2. Define a set of tasks

To evaluate the system effectively, we created a defined set of scenario-based tasks for users to complete. Before constructing these tasks we reviewed the product brief and assessed the system's playability to determine what actions it should perform versus the system's current capabilities as a prototype. This ensured that the tasks were realistic and achievable for the user interacting with the system while also testing weaknesses that were noted in the review. The tasks were designed to gradually increase in complexity, starting with elementary actions that the user could easily do (e.g. *add buildings, move buildings, and sell buildings*), which helped the user gain confidence in their abilities to use the system. Then further along in the evaluation, the tasks became more challenging, focusing on how the user could affect the gameplay beyond building the campus (e.g. *increasing/decreasing satisfaction, and reacting to in-game events*).

Each task was accompanied by follow-up questions designed to further explore the user experience and estimate how intuitive the game is. These questions were also created to encourage the use of think-aloud protocol, encouraging participants to verbalise their thoughts through the decision-making process for each task. Recording these immediate reactions during the evaluation process helped us gain better insights into usability issues.

3. Prepare an environment & Plan how to run tasks

To conduct the user evaluations effectively within our timeframe, we chose to perform them online for greater flexibility. This approach reduced the time and location restraints for both the user and evaluators, increasing participation. However, a few drawbacks were considered such as technical issues or users' lack of the necessary software to run the game. To mitigate this, we tested whether the user could open a .jar file on their devices beforehand, and if they could not we implemented a hybrid approach, where one of the evaluators met the participant in person, using their own device for the Google Meet and running the game. After we set up the environment and extenuating circumstances, the user evaluation team conducted a pilot run to refine the flow as to optimal questions alongside their order, establishing a uniform testing process.

4. Compile Our Data & Make Design Recommendations

During the user evaluations, not only did the scribe write down the answers to each question provided, they also noted any problems that the user had with the system. At the end of the evaluation, the problems would be reviewed and the user would be asked to rate the problem from 1 to 4 (***1****-Cosmetic,* ***2****-Minor,* ***3****-Major,* ***4****-Catastrophic*). This overall helped us compile the data from each of the user documents into the table below. Due to the time gaps between each user evaluation, the team decided that it would be beneficial to share feedback with the implementation team as it became available so they would have as much time as possible to address issues and carry out improvements.

2 Usability Problems Table

Usability Problem: All of these problems are based on three versions of the game, starting with the prototype, followed by ones that implemented changes to address various usability problems.	Severity Rating (**1**-Cosmetic, **2**-Minor, **3**-Major, **4**-Catastrophic)			
Prototypes **(v0.1.6, v0.2.1, v0.3.0)**	1	2	3	4
Building: User cannot undo/backtrack when placing a building, unable to change the building type, limiting flexibility				**X**
Building: Red/Green visualisation for building placement poses accessibility challenges for color blind users		**X**		
Building: The allocated building budget of £800,000 was considered too high, reducing the challenge and realism that budget is meant to add	**X**			
Building: 'Press *ESC* to cancel building placement' tip gets in the way of building placement, hindering overall playability.	**X**			
Building: As a smaller tile element, roads are difficult to place on the map.		**X**		
Events: "Roses" event lacks clarity on what the benefits and consequences are to reacting either positively or negatively.		**X**		
Events: "Roses" event lacks clarity as to whether you are given time to modify the campus before the event happens.				**X**
Events: The way events appear on screen is abrupt, distracting users.				**X**
Satisfaction: Logic behind how satisfaction rises and falls is unclear to users.				**X**
Leaderboard: When the leaderboard has no data, it lacks any sort of visual indicators such as empty spots that suggest they can be filled.		**X**		
Leaderboard: Scores are overwritten if the same name is used, so a user cannot see their improved performance overtime.			**X**	
Usability: Text size on event/tutorial pop-ups is too small, causing eye strain				**X**
Usability: Key interface elements such as the map for building placement, building counter, timers, satisfaction score are too small to attract user attention effectively.			**X**	
Usability: The pause button is in an inconvenient location (worst visual field)		**X**		
Usability: Game lacks clear direction, as objectives feel hidden to the user.				**X**
Usability: Tutorial description is vague, especially when explaining events, how to gain satisfaction, and the concept of "growing the university"			**X**	
Usability: Keyboard shortcuts are not clear or specifically mentioned, such as the 'E' & 'Q' keys to zoom in and out.			**X**	
Usability: Camera movement control to better visualise areas of the map are unintuitive and confusing, overall reducing usability.			**X**	
Usability: Lack of variety in game events and buildings with the types, decreasing engagement in the gameplay.		**X**		